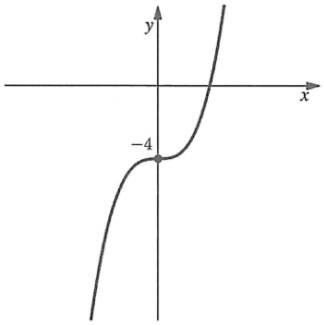
Graph Sketching: Cubic Graphs

Questions 1–16

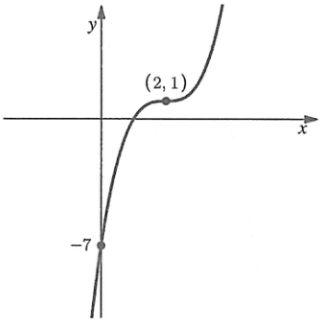
Question 1. The following graphs are all functions that can be written in the form $y = (x - p)^3 + q$. Find the equation of the curve in the form

$$y = x^3 + bx^2 + cx + d$$

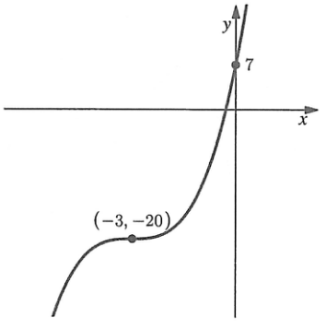
a)



b)



c)



Question 2. Sketch the following graphs:

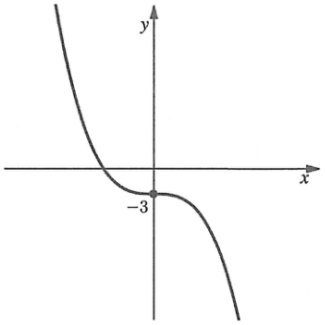
a) $y = (x + 5)(x + 3)(x + 1)$

b) $y = (x - 5)(x - 3)(x - 1)$

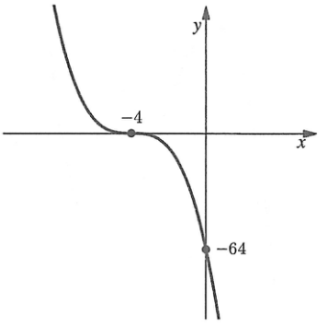
Question 3. The following graphs are all functions that can be written in the form $y = q - (x - p)^3$. Find the equation of the curve in the form

$$y = -x^3 + bx^2 + cx + d$$

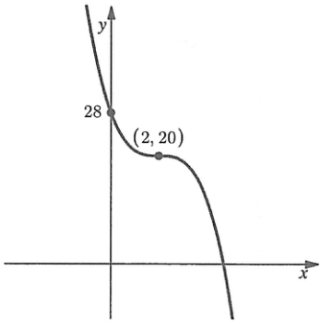
a)



b)



c)



Question 4. Sketch the following graphs:

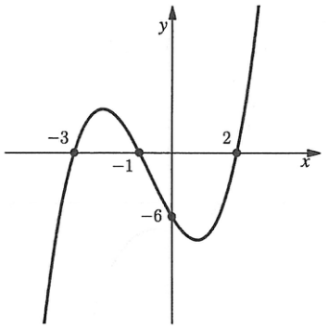
a) $y = -(x + 5)(x + 3)(x + 1)$

b) $y = -(x - 5)(x - 3)(x - 1)$

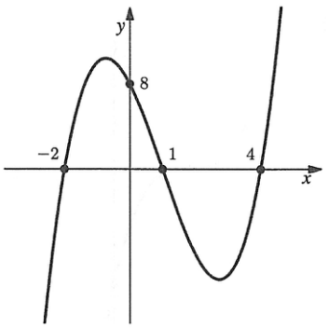
Question 5. Find the equation of each of the following graphs in the form

$$y = (x - a)(x - b)(x - c)$$

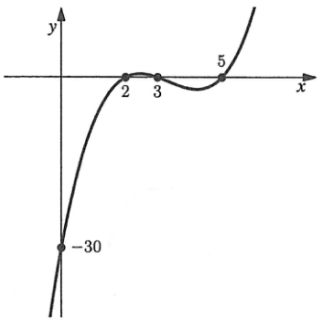
a)



b)



c)



Question 6. Sketch the following graphs:

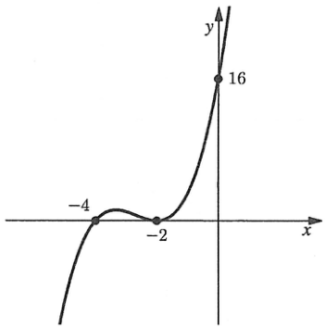
a) $y = (x + 2)(x - 2)^2$

b) $y = (x - 2)(x + 2)^2$

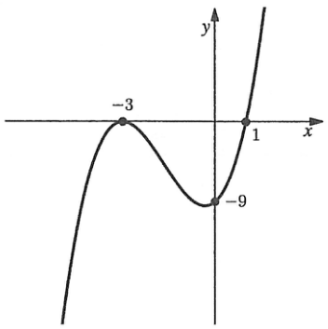
Question 7. Find the equation of each of the following graphs in the form

$$y = (x - a)(x - b)^2$$

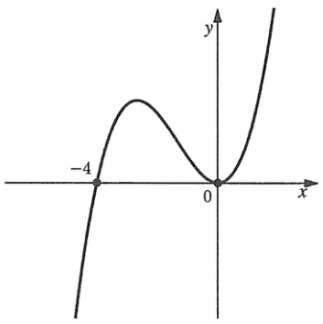
a)



b)



c)



Question 8. Sketch the following graphs:

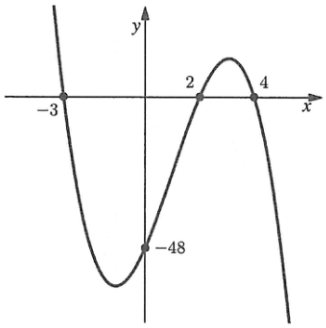
a) $y = -(x + 3)(x - 3)^2$

b) $y = (3 - x)(x + 3)^2$

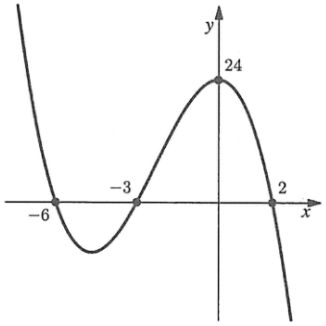
Question 9. Find the equation of each of the following graphs in the form

$$y = k(x-a)(x-b)(x-c)$$

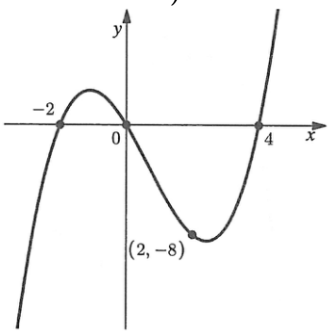
a)



b)



c)



Question 10. Sketch the following graphs:

a) $y = \frac{1}{3}(x-6)(x-3)(x+3)$

b) $y = -\frac{3}{4}(x+8)(x+4)(x-4)$

Question 11. Sketch the following graphs, showing where they intersect the coordinate axes:

a) $y = (2x + 1)(x - 2)(2x - 7)$
b) $y = (3x - 1)(2x - 3)^2$
c) $y = x(4x - 1)^2$
d) $y = x^2(4x - 1)$

Question 12. Sketch the following graphs, showing where they intersect the coordinate axes:

a) $y = -(x + 3)(2x - 1)(2x - 7)$
b) $y = (4 - x)(3x + 2)^2$
c) $y = \frac{1}{5}(x + 5)^2(5x - 4)$
d) $y = -\frac{5}{3}(x + 9)(x - 3)(x - 6)$

Question 13. Given that $k > 0$, sketch the following graphs:

a) $y = (x - k)(x - 2k)(x - 3k)$

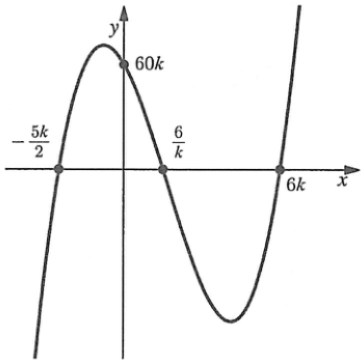
b) $y = (x + k)(x + 2k)(x - 3k)$

c) $y = (k - x)(x - 2k)(x + 3k)$

d) $y = (k - x)(2k - x)(x + 3k)$

Question 14. Given that k is a positive constant, find the equation of the following graph in the form

$$y = a(bx + c)(dx + e)(fx + g)$$



Question 15. Given that $k > 0$, sketch the following graph:

$$y = k(2k - x)(x - k)^2$$

Question 16. Given that $k > 0$, sketch the graphs of

$$y = (x - 2k)(3kx + 20)$$

and

$$y = (x - 2k)(3kx + 20)x$$

on the same axes, showing clearly where the two graphs cross the coordinate axes and each other.